

ISC Class XII Physics

Chapter 5 Magnetism and Matter

17 Numericals Answer Key

Step-by-Step ISC Board Solutions

Topics Covered

- I. Magnetic Dipole, Magnetic Moment, etc.
- II. Earth Magnetism

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Formula Sheet

Physical Constants Used in This Chapter

Permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$
$\frac{\mu_0}{4\pi}$	$= 1 \times 10^{-7} \text{ T m A}^{-1}$
Conversion	$1 \text{ G} = 10^{-4} \text{ T}$

Section I: Magnetic Dipole, Magnetic Moment, etc.

Torque on a Magnetic Dipole

$$\tau = mB \sin \theta$$

m = magnetic moment, B = magnetic field, θ = angle between m and B .

Potential Energy of a Dipole

$$U = -mB \cos \theta = -\vec{m} \cdot \vec{B}$$

Minimum (most stable) when $\theta = 0^\circ$; maximum (unstable) when $\theta = 180^\circ$.

Axial Field of a Short Bar-Magnet

$$B_{\text{axis}} = \frac{\mu_0}{4\pi} \cdot \frac{2m}{r^3}$$

r = distance from centre of magnet along its axis.

Equatorial Field of a Short Bar-Magnet

$$B_{\text{eq}} = \frac{\mu_0}{4\pi} \cdot \frac{m}{r^3}$$

r = distance from centre of magnet along the equatorial line.

Magnetic Moment of an Equivalent Current Loop

$$m = NIA$$

N = number of turns, I = current, A = area of the loop.

Section II: Earth Magnetism

Components of Earth's Magnetic Field

$$B_E = \sqrt{H^2 + V^2}, \quad \tan \delta = \frac{V}{H}$$

H = horizontal component, V = vertical component, δ = angle of dip.

Magnetic Moment of a Uniformly Magnetised Rod

$$m = M \times \text{Volume} = M(\pi r^2 L)$$

M = intensity of magnetisation, r = radius, L = length of the rod.

Permeability and Susceptibility

$$\mu = \frac{B}{H}, \quad \mu_r = \frac{\mu}{\mu_0}, \quad \chi = \mu_r - 1$$

B = magnetic field induction, H = magnetising field, μ_r = relative permeability, χ = magnetic susceptibility.

Intensity of Magnetisation

$$M = \chi H$$

χ = magnetic susceptibility, H = magnetising field.

Curie's Law (Paramagnetic Materials)

$$\chi \propto \frac{1}{T}, \quad \frac{M}{B} \propto \frac{1}{T}$$

T = absolute temperature; susceptibility (and magnetic moment per unit field) rises as T falls.

I. Magnetic Dipole, Magnetic Moment, etc.

Application Based Questions

Question 1

A small magnetic needle of magnetic moment $4.8 \times 10^{-2} \text{ J T}^{-1}$ is placed at 30° with a uniform external magnetic field $\vec{B}$ of magnitude $3.0 \times 10^{-2} \text{ T}$. What is the magnitude and direction of the torque acting on the needle?

Given: $m = 4.8 \times 10^{-2} \text{ J T}^{-1}$, $B = 3.0 \times 10^{-2} \text{ T}$, $\theta = 30^\circ$

Formula: $\tau = mB \sin \theta$

Solution:

$$\begin{aligned}\tau &= (4.8 \times 10^{-2})(3.0 \times 10^{-2}) \sin 30^\circ \\ \tau &= (4.8 \times 10^{-2})(3.0 \times 10^{-2})(0.5) = 7.2 \times 10^{-4} \text{ J}\end{aligned}$$

The torque always acts so as to reduce the angle between $\vec{m}$ and $\vec{B}$.

Final Answer

$\tau = 7.2 \times 10^{-4} \text{ J}$, direction such that it tends to bring the needle along $\vec{B}$.

Question 2

A bar-magnet placed in a uniform magnetic field of 0.3 T , with its axis at 30° to the field, experiences a torque of 0.06 N m . Find the magnetic moment of the bar-magnet.

Given: $B = 0.3 \text{ T}$, $\theta = 30^\circ$, $\tau = 0.06 \text{ N m}$

Formula: $\tau = mB \sin \theta \Rightarrow m = \frac{\tau}{B \sin \theta}$

Solution:

$$\begin{aligned}m &= \frac{0.06}{0.3 \times \sin 30^\circ} = \frac{0.06}{0.3 \times 0.5} = \frac{0.06}{0.15} \\ m &= 0.4 \text{ A m}^2\end{aligned}$$

Final Answer

$m = 0.4 \text{ A m}^2$

Understanding and Application Based Questions

Question 3

A short bar-magnet of magnetic moment 0.9 J T^{-1} placed with its axis at 45° with a uniform external magnetic field experiences a torque of magnitude 0.063 J . Find the strength of the magnetic field. What orientation of the bar-magnet corresponds to the stable equilibrium in the magnetic field?

Given: $m = 0.9 \text{ J T}^{-1}$, $\theta = 45^\circ$, $\tau = 0.063 \text{ J}$

Formula: $\tau = mB \sin \theta \Rightarrow B = \frac{\tau}{m \sin \theta}$

Solution:

$$B = \frac{0.063}{0.9 \times \sin 45^\circ} = \frac{0.063}{0.9 \times 0.7071}$$

$$B = \frac{0.063}{0.6364} \approx 0.099 \text{ T}$$

For stable equilibrium, potential energy $U = -mB \cos \theta$ must be minimum, which occurs at $\theta = 0^\circ$, i.e. the magnet's axis parallel to $\vec{B}$ (zero torque, restoring on displacement).

Final Answer

$B \approx 0.099 \text{ T}$; the orientation of zero torque (magnet parallel to $\vec{B}$) corresponds to stable equilibrium.

Common Mistake: Do not confuse the angle giving zero torque with the angle giving maximum torque (90°). Stable equilibrium is at $\theta = 0^\circ$, not at the angle stated in the question.

Question 4

The magnetic moment of a short bar-magnet is 4.0 A m^2 . Find the magnetic field at 5.0 cm from the centre of the magnet on its axis. If the magnetic field at a point situated on the equatorial line of this magnet be 0.05 T , then what is the distance of that point from the magnet?

★ Likely Exam Question

Given: $m = 4.0 \text{ A m}^2$, $r_{\text{axis}} = 5.0 \text{ cm} = 5.0 \times 10^{-2} \text{ m}$, $B_{\text{eq}} = 0.05 \text{ T}$

Formula: $B_{\text{axis}} = \frac{\mu_0}{4\pi} \cdot \frac{2m}{r^3}$, $B_{\text{eq}} = \frac{\mu_0}{4\pi} \cdot \frac{m}{r^3}$

Solution:

Part 1 — Axial field:

$$B_{\text{axis}} = (1 \times 10^{-7}) \times \frac{2 \times 4.0}{(5.0 \times 10^{-2})^3} = (1 \times 10^{-7}) \times \frac{8.0}{1.25 \times 10^{-4}}$$

$$B_{\text{axis}} = (1 \times 10^{-7}) \times (6.4 \times 10^4) = 6.4 \times 10^{-3} \text{ T}$$

Part 2 — Distance for given equatorial field:

$$0.05 = (1 \times 10^{-7}) \times \frac{4.0}{r^3} \Rightarrow r^3 = \frac{4.0 \times 10^{-7}}{0.05} = 8 \times 10^{-6} \text{ m}^3$$

$$r = \sqrt[3]{8 \times 10^{-6}} = 2.0 \times 10^{-2} \text{ m} = 2.0 \text{ cm}$$

Final Answer

$B_{\text{axis}} = 6.4 \times 10^{-3} \text{ T}$; distance on equatorial line = 2.0 cm

Common Mistake: Remember the axial field is *twice* the equatorial field at the same distance — using the equatorial formula for the axial part (or vice-versa) is a very common slip.

Question 5

Two small magnets are placed along a straight line at a distance of 20 cm from each other and their north poles are facing each other. If their magnetic moments be 0.20 and 0.40 J T^{-1} , then find the magnetic field at the middle-point between them.

Given: $d = 20 \text{ cm}$, so each magnet is at $r = 10 \text{ cm} = 0.10 \text{ m}$ from the mid-point; $m_1 = 0.20 \text{ J T}^{-1}$, $m_2 = 0.40 \text{ J T}^{-1}$

Formula: $B_{\text{axis}} = \frac{\mu_0}{4\pi} \cdot \frac{2m}{r^3}$

Solution:

The two magnets lie *along the same straight line*, so the mid-point lies on the **axis** of both magnets (not the equatorial line).

$$B_1 = (1 \times 10^{-7}) \times \frac{2 \times 0.20}{(0.10)^3} = (1 \times 10^{-7}) \times \frac{0.40}{1 \times 10^{-3}} = 4.0 \times 10^{-5} \text{ T}$$

$$B_2 = (1 \times 10^{-7}) \times \frac{2 \times 0.40}{(0.10)^3} = (1 \times 10^{-7}) \times \frac{0.80}{1 \times 10^{-3}} = 8.0 \times 10^{-5} \text{ T}$$

Since both north poles face each other (point towards the mid-point), the axial field of magnet 1 at the mid-point points away from magnet 1 (towards magnet 2), while the axial field of magnet 2 points away from magnet 2 (towards magnet 1) — the two fields point in *opposite* directions along the line, so they subtract:

$$B_{\text{net}} = B_2 - B_1 = 8.0 \times 10^{-5} - 4.0 \times 10^{-5} = 4.0 \times 10^{-5} \text{ T}$$

Final Answer

$B_{\text{net}} = 4.0 \times 10^{-5} \text{ T}$

Common Mistake: Because the two magnets lie along the same line joining them, this is an *axial*-field problem, not an equatorial one — using the equatorial formula here is a common slip. Also always check whether the two individual fields point the same way (add) or opposite ways (subtract) before combining magnitudes.

Application and Analysis Based Question

Question 6

A short bar-magnet of magnetic moment $\vec{m} = 0.32 \text{ J T}^{-1}$ is free to rotate in the plane of a uniform external magnetic field $\vec{B}$ of 0.15 T. What orientations of the magnet would correspond to its (i) stable and (ii) unstable equilibrium? What is the potential energy of the magnet in the magnetic field in each case?

★ Likely Exam Question

Given: $m = 0.32 \text{ J T}^{-1}$, $B = 0.15 \text{ T}$

Formula: $U = -mB \cos \theta$

Solution:

(i) **Stable equilibrium** ($\theta = 0^\circ$, $\vec{m} \parallel \vec{B}$):

$$U = -mB \cos 0^\circ = -mB = -(0.32)(0.15) = -0.048 \text{ J}$$

(ii) **Unstable equilibrium** ($\theta = 180^\circ$, $\vec{m}$ anti-parallel to $\vec{B}$):

$$U = -mB \cos 180^\circ = +mB = +(0.32)(0.15) = +0.048 \text{ J}$$

Final Answer

(i) $\vec{m}$ parallel to $\vec{B}$; $U = -mB = -0.048 \text{ J}$

(ii) $\vec{m}$ anti-parallel to $\vec{B}$; $U = +mB = +0.048 \text{ J}$

Analysis, Evaluate and Application Based Question

Question 7

A bar-magnet of magnetic moment 1.5 A m^2 , when suspended perpendicular to a magnetic field, is acted upon by a couple of 0.24 N-m. (a) What is the magnitude of the magnetic field? (b) What will be the change in the potential energy of the magnet in coming from this position to the position of steady equilibrium? (c) If this magnet is equivalent to a current-carrying coil of 100 turns having a surface area of 150 cm^2 , then what will be the current in the coil?

★ Likely Exam Question

Given: $m = 1.5 \text{ A m}^2$, $\theta = 90^\circ$, $\tau = 0.24 \text{ N m}$, $N = 100$, $A = 150 \text{ cm}^2 = 1.5 \times 10^{-2} \text{ m}^2$

Formula: $\tau = mB \sin \theta$, $U = -mB \cos \theta$, $m = NIA$

Solution:

(a) **Magnetic field:** At $\theta = 90^\circ$, $\sin \theta = 1$, so $\tau = mB$.

$$B = \frac{\tau}{m} = \frac{0.24}{1.5} = 0.16 \text{ T}$$

(b) **Change in potential energy:** Initial position $\theta = 90^\circ \Rightarrow U_i = -mB \cos 90^\circ = 0$.
Steady (stable) equilibrium $\theta = 0^\circ \Rightarrow U_f = -mB \cos 0^\circ = -mB$.

$$\Delta U = U_f - U_i = -mB - 0 = -(1.5)(0.16) = -0.24 \text{ J}$$

So the potential energy *decreases* by 0.24 J.

(c) **Equivalent coil current:**

$$I = \frac{m}{NA} = \frac{1.5}{100 \times 1.5 \times 10^{-2}} = \frac{1.5}{1.5} = 1.0 \text{ A}$$

Final Answer

(a) $B = 0.16 \text{ T}$ (b) $\Delta U = 0.24 \text{ J}$ (decrease) (c) $I = 1.0 \text{ A}$

Common Mistake: Convert the area from cm^2 to m^2 before using $m = NIA$ — forgetting the 10^{-4} factor is a very common slip.

II. Earth Magnetism

Understanding and Application Based Questions

Question 8

Two mutually perpendicular lines are drawn on a table. Two small magnets, whose magnetic moments are 0.108 J T^{-1} and 0.192 J T^{-1} , are placed on these lines. If the distance of the point of intersection of these lines be respectively 30 and 40 cm from these magnets, then find the resultant magnetic field at the point of intersection.

Given: $m_1 = 0.108 \text{ J T}^{-1}$, $r_1 = 30 \text{ cm} = 0.30 \text{ m}$; $m_2 = 0.192 \text{ J T}^{-1}$, $r_2 = 40 \text{ cm} = 0.40 \text{ m}$

Formula: $B_{\text{axis}} = \frac{\mu_0}{4\pi} \cdot \frac{2m}{r^3}$, $B_{\text{net}} = \sqrt{B_1^2 + B_2^2}$ (perpendicular fields)

Solution:

Since each magnet lies along its own line and the intersection point lies on that same line, the point is on the **axis** of each magnet.

$$B_1 = (1 \times 10^{-7}) \times \frac{2 \times 0.108}{(0.30)^3} = (1 \times 10^{-7}) \times \frac{0.216}{0.027} = 8.0 \times 10^{-7} \text{ T}$$

$$B_2 = (1 \times 10^{-7}) \times \frac{2 \times 0.192}{(0.40)^3} = (1 \times 10^{-7}) \times \frac{0.384}{0.064} = 6.0 \times 10^{-7} \text{ T}$$

As the two lines are mutually perpendicular, B_1 and B_2 are also perpendicular to each other, so they combine like vectors at right angles:

$$B_{\text{net}} = \sqrt{B_1^2 + B_2^2} = \sqrt{(8.0 \times 10^{-7})^2 + (6.0 \times 10^{-7})^2} = \sqrt{100} \times 10^{-7}$$

$$B_{\text{net}} = 10 \times 10^{-7} = 1.0 \times 10^{-6} \text{ T}$$

Final Answer

$$B_{\text{net}} = 1.0 \times 10^{-6} \text{ T}$$

Common Mistake: Since both fields are axial (not one axial, one equatorial), and mutually perpendicular, they must be combined using the Pythagorean rule — do not simply add or subtract the magnitudes.

Question 9

A magnetic needle free to rotate in a vertical plane parallel to the magnetic meridian has its north tip pointing down at 60° with the horizontal. The horizontal component of earth's magnetic field is $0.3 \times 10^{-4} \text{ Wb/m}^2$. Find the vertical component and total

magnetic field of earth.

Given: $\delta = 60^\circ$, $H = 0.3 \times 10^{-4} \text{ Wb/m}^2$

Formula: $\tan \delta = \frac{V}{H}$, $B_E = \frac{H}{\cos \delta}$

Solution:

$$V = H \tan \delta = (0.3 \times 10^{-4}) \times \tan 60^\circ = (0.3 \times 10^{-4}) \times 1.732$$

$$V = 0.52 \times 10^{-4} \text{ Wb/m}^2$$

$$B_E = \frac{H}{\cos 60^\circ} = \frac{0.3 \times 10^{-4}}{0.5} = 0.60 \times 10^{-4} \text{ Wb/m}^2$$

Final Answer

$$V = 0.52 \times 10^{-4} \text{ Wb/m}^2, \quad B_E = 0.60 \times 10^{-4} \text{ Wb/m}^2$$

Question 10

A magnetic needle free to rotate about the vertical direction (compass) points 3.5° west of the geographic north. Another magnetic needle free to rotate in a vertical plane parallel to the magnetic meridian (dip needle) has its north tip pointing down at 18° with the horizontal. The horizontal component of earth's magnetic field at the place is 0.40 G. Find the magnitude and direction of the earth's magnetic field $\vec{B}_E$ at the place. ($\tan 18^\circ = 0.325$)

★ Likely Exam Question

Given: Declination = 3.5° W, Dip $\delta = 18^\circ$, $H = 0.40 \text{ G}$, $\tan 18^\circ = 0.325$

Formula: $B_E = \frac{H}{\cos \delta}$

Solution:

$$\cos 18^\circ \approx 0.9511$$

$$B_E = \frac{0.40}{0.9511} \approx 0.42 \text{ G}$$

The compass reading fixes the *horizontal* direction of the field (3.5° west of geographic north), while the dip needle fixes how far $\vec{B}_E$ tilts below the horizontal in that vertical plane (18° , north tip down — consistent with a place in the northern magnetic hemisphere).

Final Answer

$B_E \approx 0.42$ G, lying in a vertical plane 3.5° west of the geographic meridian, making an angle of 18° downward with the horizontal.

Common Mistake: The angle of dip is measured from the *horizontal*, so use $\cos \delta$ (not $\sin \delta$) to relate H and B_E .

Question 11

A cylindrical rod magnet has a length of 5.0 cm and a diameter of 1.0 cm. It has a uniform magnetisation of 5.3×10^3 A m⁻¹. Find its magnetic moment.

Given: $L = 5.0$ cm $= 5.0 \times 10^{-2}$ m, diameter = 1.0 cm $\Rightarrow r = 0.5$ cm $= 0.5 \times 10^{-2}$ m, $M = 5.3 \times 10^3$ A m⁻¹

Formula: $m = M \times \text{Volume} = M(\pi r^2 L)$

Solution:

$$\begin{aligned}\text{Volume} &= \pi r^2 L = \pi \times (0.5 \times 10^{-2})^2 \times (5.0 \times 10^{-2}) \\ &= \pi \times (2.5 \times 10^{-5}) \times (5.0 \times 10^{-2}) = \pi \times 1.25 \times 10^{-6} \approx 3.93 \times 10^{-6} \text{ m}^3 \\ m &= (5.3 \times 10^3) \times (3.93 \times 10^{-6}) \approx 20.8 \times 10^{-3} \text{ A m}^2\end{aligned}$$

Final Answer

$$m \approx 20.8 \times 10^{-3} \text{ A m}^2$$

Common Mistake: Use the *radius* (half the diameter), not the diameter itself, in the volume formula.

Application Based Questions**Question 12**

The horizontal component of earth's magnetic field at a place is $\sqrt{3}$ times the vertical component. What is angle of dip at that place?

Given: $H = \sqrt{3} V$

Formula: $\tan \delta = \frac{V}{H}$

Solution:

$$\tan \delta = \frac{V}{\sqrt{3} V} = \frac{1}{\sqrt{3}} \Rightarrow \delta = 30^\circ$$

Final Answer

$$\delta = 30^\circ$$

Question 13

If the ratio of the horizontal component of earth's magnetic field to the resultant magnetic field at a place is $1/\sqrt{2}$, what is the angle of dip at that place?

Given: $\frac{H}{B_{\text{net}}} = \frac{1}{\sqrt{2}}$

Formula: $\cos \delta = \frac{H}{B_{\text{net}}}$

Solution:

$$\cos \delta = \frac{1}{\sqrt{2}} \Rightarrow \delta = 45^\circ$$

Final Answer

$$\delta = 45^\circ$$

Question 14

Relative permeability of iron is 5500. Find its magnetic susceptibility.

Given: $\mu_r = 5500$

Formula: $\chi = \mu_r - 1$

Solution:

$$\chi = 5500 - 1 = 5499$$

Final Answer

$$\chi = 5499$$

Application and Analysis Based Questions**Question 15**

The magnetic field induction in a material placed in a magnetising field of 2000 A m^{-1} is $1.0 \text{ N A}^{-1}\text{m}^{-1}$. Find (i) magnetic permeability, (ii) relative permeability, (iii) magnetic susceptibility and (iv) intensity of magnetisation for the material.

★ Likely Exam Question

Given: $H = 2000 \text{ A m}^{-1}$, $B = 1.0 \text{ N A}^{-1}\text{m}^{-1} = 1.0 \text{ T}$

Formula: $\mu = \frac{B}{H}$, $\mu_r = \frac{\mu}{\mu_0}$, $\chi = \mu_r - 1$, $M = \chi H$

Solution:

(i) **Permeability:**

$$\mu = \frac{B}{H} = \frac{1.0}{2000} = 5 \times 10^{-4} \text{ N A}^{-2}$$

(ii) **Relative permeability:**

$$\mu_r = \frac{\mu}{\mu_0} = \frac{5 \times 10^{-4}}{4\pi \times 10^{-7}} = \frac{5 \times 10^{-4}}{1.2566 \times 10^{-6}} \approx 398$$

(iii) **Magnetic susceptibility:**

$$\chi = \mu_r - 1 = 398 - 1 = 397$$

(iv) **Intensity of magnetisation:**

$$M = \chi H = 397 \times 2000 = 7.94 \times 10^5 \text{ A m}^{-1}$$

Final Answer

(i) $\mu = 5 \times 10^{-4} \text{ N A}^{-2}$ (ii) $\mu_r \approx 398$ (iii) $\chi \approx 397$ (iv) $M \approx 7.94 \times 10^5 \text{ A m}^{-1}$

Common Mistake: $1 \text{ N A}^{-1}\text{m}^{-1}$ is the same as 1 T — do not treat it as a different unit requiring conversion.

Question 16

A paramagnetic salt has a magnetic moment of 5.0 A m^2 at a temperature of 4 K in a magnetic field of 1.0 T . Find its magnetic moment at a temperature of 3 K in a magnetic field of 1.5 T .

Given: $M_1 = 5.0 \text{ A m}^2$, $T_1 = 4 \text{ K}$, $B_1 = 1.0 \text{ T}$; $T_2 = 3 \text{ K}$, $B_2 = 1.5 \text{ T}$

Formula: $\frac{M}{B} \propto \frac{1}{T} \Rightarrow \frac{M_1}{B_1 T_1^{-1}} = \frac{M_2}{B_2 T_2^{-1}} \Rightarrow M_2 = M_1 \times \frac{B_2}{B_1} \times \frac{T_1}{T_2}$

Solution:

$$M_2 = 5.0 \times \frac{1.5}{1.0} \times \frac{4}{3} = 5.0 \times 1.5 \times 1.333$$

$$M_2 = 10.0 \text{ A m}^2$$

Final Answer

$$M_2 = 10 \text{ A m}^2$$

Common Mistake: By Curie's Law, magnetic moment is directly proportional to B but inversely proportional to T — do not multiply by T_2/T_1 instead of T_1/T_2 .

Question 17

The magnetic susceptibility of magnesium at 300 K is 1.2×10^5 . At what temperature will its magnetic susceptibility become 1.44×10^5 ?

Given: $\chi_1 = 1.2 \times 10^5$ at $T_1 = 300 \text{ K}$; $\chi_2 = 1.44 \times 10^5$

Formula: $\chi \propto \frac{1}{T} \Rightarrow \chi_1 T_1 = \chi_2 T_2$

Solution:

$$T_2 = \frac{\chi_1 T_1}{\chi_2} = \frac{(1.2 \times 10^5) \times 300}{1.44 \times 10^5}$$
$$T_2 = \frac{3.6 \times 10^7}{1.44 \times 10^5} = 250 \text{ K}$$

Final Answer

$$T_2 = 250 \text{ K}$$

Common Mistake: Since susceptibility *increases* as temperature falls, the required temperature must come out lower than 300 K — a positive result greater than 300 K signals a sign/inversion error in setting up the ratio.